

SLR USING Least square

A) Procedure

B) Load the California Housing dataset.

C) Identify the variables required for SLR:

- Independent variable X: `TotalRooms` (Total rooms)
- Dependent variable Y: `MedHouseVal` (Median House Value)

D) Inspect the dataset for missing, duplicate, or invalid values.

E) Calculate the slope m and the intercept b using the least square equation:

F) Construct the regression equation:

$$Y^{\wedge}=mX+b$$

G) Use the regression equation to predict the target values for the test dataset.

H) Calculate the Mean Squared Error (MSE) and the coefficient of determination R^2 to measure the goodness of fit.

I) Plot the actual data points and regression line.

SLR GRADIENT DESCENT

B) Procedure

1. **Load the California Housing dataset** and select `TotalRooms` as X and `MedHouseVal` as Y.
2. **Inspect and clean the data** by checking missing and invalid values.
3. **Split the dataset** into training and testing sets.
4. **Standardize the input feature** using the mean and standard deviation of the training data.
5. **Initialize the parameters:**
 - Slope $m=0$
 - Intercept $c=0$
6. **Choose the hyperparameters:**
 - Learning rate $\alpha =0.01$
 - Number of iterations/epochs= 100
7. **Calculate the predicted values**
8. **Calculate the Mean Squared Error cost:**

$$J(m,c)=1/n\sum(Y_i-Y^{\wedge}_i)^2$$

11. **Calculate the gradient with respect to slope:**

$$\partial m/\partial J=-2/n\sum X_i(Y_i-Y^{\wedge}_i)$$

12. **Calculate the gradient with respect to intercept:**

$$\partial b/\partial J=-2/n\sum(Y_i-Y^{\wedge}_i)$$

13. **Update the slope:**

$$m = m - \alpha [\partial m / \partial J]$$

14. Update the intercept:

$$b = b - \alpha [\partial b / \partial J]$$

15. **Repeat Steps 8–14** for the specified number of epochs or until the cost function converges.
16. **Plot the cost versus epochs** to verify convergence.
17. **Use the final values of m and b** to predict the house values for the test dataset.
18. **Calculate MSE and R2** for the Gradient Descent model.

SLR using sci-kit learn

C) PROCEDURE

1. Import the required libraries

- Import NumPy and Pandas for data handling.
- Import `fetch_california_housing` from Scikit-learn.
- Import `train_test_split` for splitting the dataset.
- Import `LinearRegression` for building the SLR model.
- Import `mean_squared_error` and `r2_score` for evaluation.
- Import Matplotlib for visualization.

2. Load the California Housing dataset

- Load the dataset using Scikit-learn's `fetch_california_housing()` function.

3. Create a DataFrame

- Convert the dataset features into a Pandas DataFrame.
- Assign the corresponding feature names to the columns.
- Store the target variable separately.

4. Explore the dataset

- Display the first few records.
- Check the shape of the dataset.
- Examine the available features.
- Check for missing values.

5. Select variables for SLR

- Select **AveRooms(Total rooms)** as the independent variable X .
- Select **MedHouseVal (Median House Value)** as the dependent variable Y .

$X = \text{AveRooms}$ $Y = \text{MedHouseVal}$

6. Visualize the relationship

- Create a scatter plot between `AveRooms` and `MedHouseVal`.
- Observe whether a linear relationship exists between the variables.

7. Split the dataset

- Divide the data into training and testing sets.
- Use **80% for training** and **20% for testing**.
- Set `random_state=42` to obtain reproducible results.

8. Create the SLR model

- Create an instance of the Scikit-learn `LinearRegression` model.

9. Train the model

- Fit the model using the training data.

10. Obtain the model parameters

- Determine the **coefficient/slope m** .
- Determine the **intercept b** .

11. Predict the test data

- Use the trained model to predict `MedHouseVal` for the test samples.

12. Evaluate the model using MSE and R2

13. Visualize the regression line

- Plot the actual test observations.
- Plot the fitted regression line.
- Add suitable labels, title, and legend.